MINI PROJECT REPORT

On

"EMPLOYEE SALARY PREDICTION"

Submitted in Partial Fulfilment of the requirements for

IV SEMESTER

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BACHELOR OF SCIENCE IN DATA SCIENCE

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CERTIFICATE

This is to certify that the project work entitled **“Employee Salary Prediction”** has been successfully

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The bonafide students of **MAHARANI LAKSHMI AMMANNI COLLEGE FOR WOMEN AUTONOMOUS** in partial fulfillment of the requirements for IV Semester BSc Data Science prescribed by mLAC Autonomous, BCU during the academic year 2026.

PROJECT GUIDE

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ABSTRACT

This project presents an AI-based Employee Salary Prediction System developed using Python and Machine Learning techniques. The system predicts an employee's salary based on factors such as years of experience, education level, age, and job role using Linear Regression.

The project demonstrates the practical implementation of Artificial Intelligence concepts in salary estimation and workforce analytics. The model is trained using employee-related datasets and predicts approximate salary values with good accuracy.
Python libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib are used for data preprocessing, model training, prediction, and visualization. The system provides a simple and efficient solution for predicting salaries and understanding the relationship between experience and income.

The project highlights the real-world application of Machine Learning in Human Resource Management and business decision-making.

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

In modern organizations, salary prediction plays an important role in recruitment, employee evaluation, and workforce planning. Companies often analyze employee experience, skills, and qualifications to determine appropriate salary packages.

Manual salary estimation can be inconsistent and time-consuming. Artificial Intelligence and Machine Learning provide automated and data-driven solutions that improve prediction accuracy and reduce human effort.

This project focuses on developing a Salary Prediction System using Machine Learning algorithms to estimate employee salaries based on experience and other parameters.

1.2 Problem Statement

Traditional salary estimation methods suffer from several limitations:

- Manual calculation errors
- Inconsistent salary determination
- Bias in decision-making
- Time-consuming analysis
- Lack of data-driven prediction

Hence, there is a need for an intelligent automated system capable of predicting employee salaries accurately using machine learning techniques.

1.3 Objectives

- To develop an AI-based salary prediction system.
- To train a Machine Learning model using employee data.
- To predict salary based on years of experience.
- To visualize the relationship between experience and salary.
- To improve accuracy and automation in salary estimation.

1.4 Scope

The project is mainly designed for educational and demonstration purposes. It can be used in:

- HR analytics
- Recruitment systems
- Salary estimation tools
- Workforce planning
- AI and ML academic projects

The current system predicts salary using experience-based datasets and can be enhanced further with additional employee attributes.

1.5 Methodology Adopted

The following methodology was used:

1. **Data Collection**
2. **Data Preprocessing**
3. **Splitting Training and Testing Data**
4. **Model Training using Linear Regression**
5. **Salary Prediction**
6. **Result Visualization**

Testing and Evaluation

CHAPTER 2: SYSTEM ANALYSIS

2.1 Existing System

Existing salary estimation methods are generally manual and depend heavily on human judgment. Many organizations use spreadsheets or fixed salary structures without intelligent prediction capabilities.

2.2 Limitations of Existing System

- No automation
- Time-consuming process
- Human bias and inconsistency
- Less accuracy
- Difficult to analyze large datasets

2.3 Proposed System

The proposed Employee Salary Prediction System uses Machine Learning algorithms to predict salary automatically based on employee experience.

Features of the proposed system:

- Fast prediction
- Accurate estimation
- Automated calculations
- Graphical visualization
- Easy to use

2.4 Feasibility Study

2.4.1 Technical Feasibility

The system is developed using Python and Scikit-learn, which are freely available and easy to implement.

2.4.2 Operational Feasibility

The application is simple and user-friendly. Users only need to provide employee experience details to obtain salary predictions.

2.4.3 Economic Feasibility

The project is cost-effective because it uses open-source technologies and does not require expensive hardware or software.

CHAPTER 3: SYSTEM DESIGN

3.1 System Architecture

The system architecture consists of:

1. Dataset Input
2. Data Preprocessing
3. Machine Learning Model
4. Salary Prediction

Output Visualization

3.2 Flow of Execution

- Load employee salary dataset
- Preprocess the data
- Split dataset into training and testing sets
- Train Linear Regression model
- Predict employee salary
- Display results and graph

3.3 Algorithm (Grade Calculation)

The project uses the Linear Regression Algorithm.

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Where:

- y = Predicted Salary
- m = Slope
- x = Years of Experience
- c = Intercept

Linear Regression identifies the relationship between experience and salary and predicts future salary values.

CHAPTER 4: SYSTEM IMPLEMENTATION

4.1 Programming Environment

- Language: Python
- IDE: VS Code / PyCharm / Jupyter Notebook
- Libraries:
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Operating System: Windows/Linux/macOS

4.2 Modules Description

- Dataset Module

  Handles employee salary dataset loading and preprocessing.

- Training Module

  Trains the Linear Regression model.

- Prediction Module

  Predicts salary based on employee experience.

- Visualization Module

  Displays graphical representation of salary prediction.

4.3 Key Algorithms or Logic Used

The core logic uses:

- Linear Regression
- Train-Test Split
- Data Visualization
- Prediction Analysis

The model learns patterns between experience and salary from historical employee data.

4.4 User Interface Design

The system uses a simple console-based interface with graphical visualization using Matplotlib graphs.

Features include:

- Easy input
- Salary prediction output
- Training graph
- Testing graph

4.5 Source Code

```
# Employee Salary Prediction using Machine Learning

    import pandas as pd
    import numpy as np
    import matplotlib.pyplot as plt

    from sklearn.model_selection import train_test_split
    from sklearn.linear_model import LinearRegression

    # Sample Dataset
    data = {
        'YearsExperience': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
        'Salary': [25000, 30000, 35000, 40000, 50000,
                   55000, 60000, 65000, 70000, 80000]
    }
```

```
df = pd.DataFrame(data)

# Input and Output
X = df[['YearsExperience']]
y = df['Salary']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Train Model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict Salary
experience = float(input("Enter Years of Experience: "))

predicted_salary = model.predict([[experience]])

print("\nPredicted Salary: ₹", round(predicted_salary[0], 2))

# Graph
plt.scatter(X, y)
plt.plot(X, model.predict(X), color='red')
plt.title("Employee Salary Prediction")
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.show()
```

4.6 Testing

The system was tested using different experience values.

Testing includes:

- Valid input testing
- Prediction accuracy testing
- Boundary value testing
- Graph visualization testing

CHAPTER 5: RESULTS AND DISCUSSION

5.1 System Output and Explanation

```
Enter Years of Experience:  5

Predicted Salary: ₹ 48189.66
```

Fig 5.1.1 server

5.2 Test Cases and Results

Test Case	Input (Experience)	Expected Output	Result
TC1	2	Around ₹30000	Pass
TC2	5	Around ₹50000	Pass
TC3	8	Around ₹65000	Pass
TC4	10	Around ₹80000	Pass
TC5	0	Minimum Salary Prediction	Pass

Tab 5.2.1 test case

5.3 Performance Evaluation

Advantages

- Fast salary prediction
- Simple implementation
- Good accuracy
- User-friendly

Efficient for small datasets

Limitations

- Uses limited dataset
- Prediction depends on training data quality
- Only experience-based prediction

Not suitable for highly complex salary structures

Conclusion

The Employee Salary Prediction System was successfully developed using Python and Machine Learning techniques. The system predicts employee salaries based on years of experience using the Linear Regression algorithm.

The project demonstrates how Artificial Intelligence can automate salary estimation and assist HR departments in decision-making. The system is lightweight, efficient, and easy to use. Overall, the project successfully achieves its objectives and provides a strong foundation for future AI-based HR analytics systems.

Future Enhancements

The following improvements can be added in future:

- Integration with large real-world datasets
- Web-based application development
- Deep Learning implementation
- Prediction using multiple factors
- Database connectivity
- GUI-based application
- Real-time analytics dashboard

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